TEXTBOOK OF AGRICULTURAL STATISTICS · CHAPTER 3

5-MINUTE REVISION

Graphical Representation

From the chapter's own revision points, formulae and questions.

Selected from Chapter 3 · Chapter Summary in the textbook.

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STAT 3202 · Chapter revision

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Quick revision · 1

- Histogram → continuous quantitative data; adjacent bars.
- Bar chart → categorical data; bars separated by gaps.
- Equal class width → height represents frequency.
- Unequal class width → height represents frequency density.
- Less than ogive → upper class boundaries + LCF.

Quick revision · 2

- Greater than ogive → lower class boundaries + GCF.
- Intersection of two ogives → median.
- Frequency polygon → class midpoints + frequencies.
- Stem-and-leaf plot → retains individual observations.
- Simple bar chart → one value for each category.

Quick revision · 3

- Component bar chart → shows components of each category.
- Grouped bar chart → compares multiple values within categories.
- Pie chart → sectors represent parts of a whole.
- Boxplot → five-number summary and spread of data.
- IQR → $Q_3 - Q_1$.

Quick revision · 4

- Outliers in Boxplot → observations below $Q_1 - 1.5IQR$ or above $Q_3 + 1.5IQR$.

Important formulae · 1

Frequency density:

$$\text{Frequency Density} = \frac{\text{Class Frequency}}{\text{Class Width}}$$

Angle of a sector in a pie chart:

$$\text{Sector angle} = \frac{\text{Value of category}}{\text{Sum of category values}} \times 360^\circ$$

Interquartile range:

$$IQR = Q_3 - Q_1$$

Important formulae · 2

Lower limit for identifying outliers in a boxplot:

$$Q_1 - 1.5 \times IQR$$

Upper limit for identifying outliers in a boxplot:

$$Q_3 + 1.5 \times IQR$$

Self-check from the chapter

1. A graph representing a continuous frequency distribution by adjoining rectangles is called a _____. **[Histogram]**
2. For unequal class widths, histogram heights should represent frequency _____. **[Density]**
3. A graph of cumulative frequencies is called an _____. **[Ogive]**
4. The intersection of the less-than and greater-than ogives provides an estimate of the _____. **[Median]**